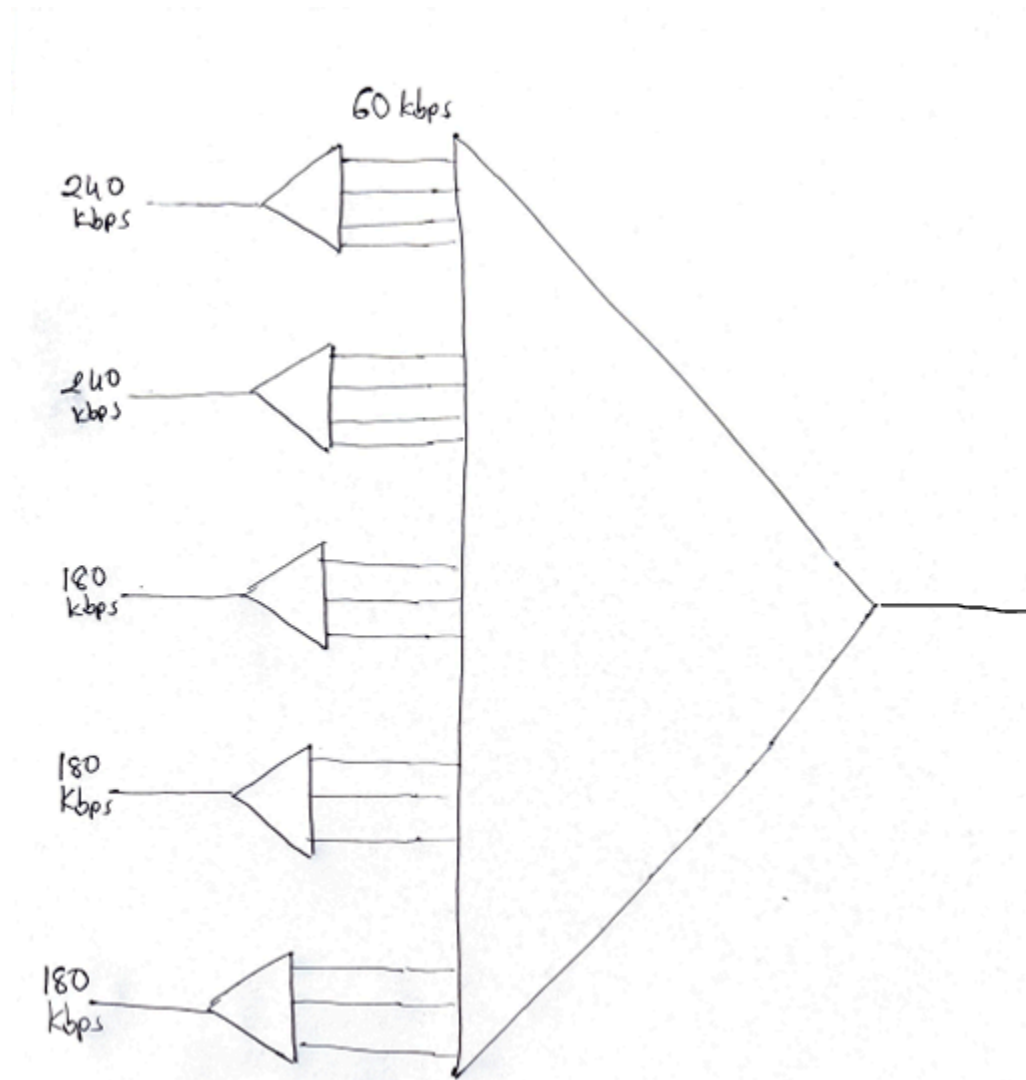


Q3. a) Since the data rates of the sources are not equal, we will use multiple-slot multiplexing. The GCD of all data rates (180 and 240) is 60, so we will allocate 60 kbps to each slot.



b) There are 17 channels in total, each with a data rate of 60 kbps.

Therefore, input data rate = 60 kbps

$$\text{Input bit duration} = \frac{1}{\text{input data rate}} = 1.667 \times 10^{-5} \text{ seconds}$$

c) Input slot duration = input bit duration $\times$ number of bits in one slot (multiplexing unit)
 $= 1.667 \times 10^{-5} \times 2 = 3.333 \times 10^{-5} \text{ seconds}$

Frame duration = input slot duration = $3.333 \times 10^{-5} \text{ seconds}$

$$\text{Frame rate} = \frac{1}{\text{frame duration}} = 30000 \text{ fps}$$

Frame size = number of channels $\times$ interleaved bits (multiplexing unit) + sync bits
 $= 17 \times 2 + 1 = 35 \text{ bits}$

d) Output rate = frame rate $\times$ frame size = 1050000 bps = 1.05 Mbps

$$\text{Output bit duration} = \frac{1}{\text{output rate}} = 9.524 \times 10^{-7} \text{ seconds}$$

$$\text{e) Output slot duration} = \frac{\text{input slot duration}}{\text{number of channels}} = 1.961 \times 10^{-6} \text{ seconds}$$

$$\begin{aligned} \text{Alternatively, output slot duration} &= \text{output bit duration} \times \text{number of bits in one slot} \\ &= 9.524 \times 10^{-7} \times 2 = 1.905 \times 10^{-7} \text{ seconds} \end{aligned}$$

[Both of these answers are acceptable]

Q4. For ease of calculation, we shall convert everything to bits.

- a) Input data rate = 1000 pages per 5 seconds
= 200 pages/second
= 200 $\times$ 20 lines/second
= 200 $\times$ 20 $\times$ 40 characters/second
= 200 $\times$ 20 $\times$ 40 $\times$ 8 bits/second
= 1280000 bps
= 1.28 Mbps

$$\text{Input bit duration} = \frac{1}{\text{input data rate}} = 7.8125 \times 10^{-7} \text{ seconds}$$

- b) Multiplexing unit = 5 lines = 5 $\times$ 40 characters = 5 $\times$ 40 $\times$ 8 bits = 1600 bits

$$\begin{aligned} \text{Input slot duration} &= \text{input bit duration} \times \text{number of bits in one slot (multiplexing unit)} \\ &= 7.8125 \times 10^{-7} \times 1600 = 1.25 \times 10^{-3} \text{ seconds} \end{aligned}$$

$$\text{Frame duration} = \text{input slot duration} = 1.25 \times 10^{-3} \text{ seconds}$$

$$\text{Frame rate} = \frac{1}{\text{frame duration}} = 800 \text{ fps}$$

$$\begin{aligned} \text{Frame size} &= \text{number of channels} \times \text{interleaved bits (multiplexing unit)} + \text{sync bits} \\ &= 5 \times 1600 + 5 = 8005 \text{ bits} \end{aligned}$$

- c) Output data rate = frame rate $\times$ frame size = 800 $\times$ 8005 = 6404000 bps = 6.404 Mbps

$$\text{Output bit duration} = \frac{1}{\text{output rate}} = 1.562 \times 10^{-7} \text{ seconds}$$